



REVIEW

How do mRNA vaccines work?

REFERENCES	TOKENS	VERIFICATION	COMPILED
47 verified	≈9.6k	Crossref	27 Aug 2026

ABSTRACT

Messenger RNA vaccines use a lipid nanoparticle to deliver a short-lived genetic recipe for a pathogen protein, not the pathogen itself. Cells translate that recipe in the cytoplasm, and the resulting antigen trains neutralizing antibodies, memory B cells, helper T cells and killer T cells. The same formulation must balance two competing requirements: it must protect and express the RNA while provoking enough innate signalling to build immunity without suppressing translation. This balance enabled high initial efficacy against ancestral SARS-CoV-2 and protection against respiratory syncytial virus, but it does not make performance uniform across pathogens or immune contexts. Protection against infection wanes faster than protection against severe disease, variants can escape a previously matched antigen, and intramuscular vaccination produces weaker airway than blood immunity. Safety surveillance shows that serious events are uncommon, while myocarditis is a real, concentrated risk that varies by age, sex, dose and product. Manufacturing remains rapid and programmable but formulation-specific, and newer self-amplifying and circular RNA designs differ sharply in clinical maturity. The platform is therefore best understood as a tunable delivery-and-immunity system rather than a single fixed vaccine technology.

01 Introduction

Messenger RNA (mRNA) vaccination separates the information that defines an antigen from the machinery that manufactures and delivers it. The sequence can be redesigned while much of the production platform remains stable, but the biological result depends on more than sequence. Delivery into the cytoplasm, innate sensing, lymph-node organization, the anatomical site of immunity and the match between antigen and circulating pathogen all shape protection.

This creates the central tension of the platform: programmability makes the message easy to change, yet formulation and immune context determine what that message can accomplish. The sections below follow that tension from lipid-particle entry and RNA chemistry through immune memory, clinical effectiveness, waning, safety, manufacturing and next-generation RNA formats.

02 Cytoplasmic delivery makes the antigen programmable

The payload is mRNA enclosed in a lipid nanoparticle (LNP). The particle protects the RNA and releases it into the cytoplasm, where ribosomes translate it; the RNA neither enters the nucleus nor becomes DNA.^{1,2,3} Ionizable lipids bind RNA during particle formation but are closer to neutral in the body. Release from the endosome nevertheless remains inefficient, and its mechanism is still disputed.^{4,5}

What the cell makes is defined by the encoded sequence rather than by a generic vaccine product. COVID-19 vaccines encoded prefusion-stabilized spike, building on earlier

coronavirus structural work, so a validated production platform could accept a redesigned sequence.^{6,1} Programmability begins with the message, but it only becomes useful when the carrier delivers enough intact RNA.

03 RNA chemistry balances translation and innate alarm

Unmodified RNA activates innate sensors. Substituting modified nucleosides reduced signalling through Toll-like receptors 3, 7 and 8 and reduced cytokine release from human dendritic cells. Removing double-stranded RNA contaminants then eliminated detectable interferon and cytokine induction in tested cells and increased translation **10- to 1,000-fold**.^{7,8} The message is therefore engineered not only for what it encodes, but also for how conspicuously the innate immune system sees it.

The carrier supplies part of the alarm that modified RNA suppresses. In mice, one LNP formulation acted as an adjuvant through its ionizable lipid and interleukin-6. Another mouse study linked the RNA-sensing protein MDA5 and its type-I-interferon signal to killer-T-cell expansion while finding several proposed pathways dispensable. These mechanisms may not generalize across LNP formulations or from mice to humans.^{9,10,11,12}

Chemical tuning also creates trade-offs. N1-methylpseudouridine reduced innate recognition but enabled +1 ribosomal frameshifting and immune recognition of off-target products in mice and people. Recoding synonymous codons lowered the signal, and no adverse clinical outcome was established.¹³ The same design choices that improve translation can therefore introduce new molecular effects that require direct measurement.



04 Lymph nodes turn a brief signal into memory

Early human trials found neutralizing antibodies together with antiviral type-1 helper and killer T-cell responses. Killer T cells expanded after the first dose, before neutralizing antibodies had fully matured.^{14,15} Serial lymph-node sampling then detected spike-specific germinal-centre B cells for at least **12 weeks** and T-follicular-helper cells for about **6 months**. Across 1,540 clones, somatic mutation rose **3.5-fold**, accompanied by affinity-matured memory B cells and marrow plasma cells, although these cohorts were small.^{16,17,18}

The anatomical distribution of that memory is uneven. Systemic vaccination produced strong blood memory but weaker airway immunity: vaccinated participants had lower bronchoalveolar neutralization and no detectable spike-specific B or T cells there, unlike convalescents. This small study helps explain breakthrough infection but does not quantify how much the airway gap contributes.¹⁹ The delivery-to-memory pathway and the boundary between human and mouse evidence are summarized in Figure 1. Durable systemic memory is real, but it does not guarantee equally durable protection at every anatomical site.

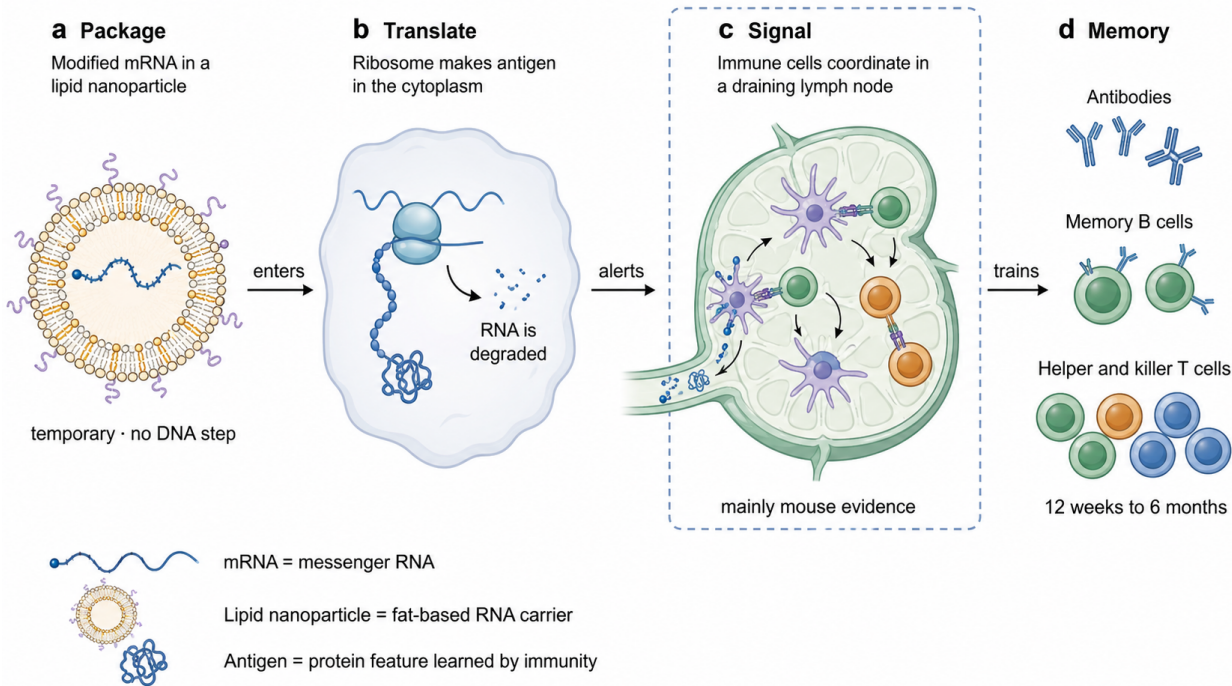


Figure 1. A temporary RNA recipe builds immune memory. An LNP delivers modified mRNA, ribosomes make antigen in the cytoplasm, immune cells co-ordinate in a draining lymph node, and antibodies plus memory B and T cells remain. Delivery and persistent human lymph-node responses are observed; the detailed sensing pathway and LNP adjuvant effect remain formulation-specific and rely heavily on mouse experiments.^{1,12,9,10,18}

05 Clinical performance depends on the target

VACCINE	EVIDENCE	MAIN RESULT	IMPORTANT BOUNDARY
BNT162b2	Placebo-controlled randomized trial, n=43,448	95.0% efficacy against symptomatic COVID-19	Ancestral-virus era ²⁰
mRNA-1273	Placebo-controlled randomized trial, n=30,420	94.1% efficacy; 30 severe cases, all placebo	Ancestral-virus era ²¹
mRNA-1345 RSV	Placebo-controlled randomized trial, n=35,541, age ≥60	83.7% against lower-respiratory disease with ≥2 signs	Median follow-up 112 days ²²
mRNA-1010 influenza	Two active-controlled phase-3 trials, >14,000	Stronger A but weaker B profile than licensed vaccine	Original formulation; further optimization needed ²³

The clearest early COVID-19 results showed what a well-matched antigen and formulation could achieve. Before major immune-escape variants, pivotal placebo-controlled trials reported **95.0% efficacy** (95% credible interval 90.3–97.6; n=43,448) for BNT162b2 and **94.1% efficacy** (95% confidence interval (CI) 89.3–96.8; n=30,420) for mRNA-1273 against symptomatic COVID-19.^{20,21} Neutralizing antibodies tracked protection: estimated mRNA-1273

efficacy rose from 78% (95% CI 54–89) at a neutralization titre of 10 to 96% (94–98) at 1,000, while a 24-study model linked variant neutralization loss to lower protection. Correlation does not establish that antibodies mediate every component of protection.^{24,25} Success extended to respiratory syncytial virus (RSV), where the platform protected older adults from disease and day-29 neutralization predicted lower risk. Influenza trials



were less uniform, producing stronger influenza-A but weaker influenza-B responses than a licensed vaccine.^{22,26,23} The platform is reusable; its effectiveness is not interchangeable across antigens.

06 Protection wanes fastest against infection

The same context dependence appears over time. Across 78 estimates, effectiveness fell **21.0 percentage points** (95% CI 13.9–29.8) against infection from month 1 to month 6, compared with **10.0 points** (6.1–15.4) against severe disease. Only 3 of 18 included studies had low overall risk of bias, and all predated widespread Omicron.²⁷ Updated vaccines restored protection without ending the pattern. A 28-study synthesis found that bivalent boosters added **28–31% relative effectiveness** against infection and **58–62%** against severe outcomes. For XBB.1.5 vaccines, estimates fell from month 1 to month 5 from 52.9% to 26.7% against infection, 64.4% to 52.3% against hospitalization and 77.3% to 69.4% against death.^{28,29}

AVERAGE DECLINE · MONTH 1 TO MONTH 6



Estimates also depend on study population and design. A target-trial emulation of 587,137 matched pairs, mostly older men, estimated lower 120-day effectiveness of 3.15% against infection, 25.25% against hospitalization and 30.25% against death. Residual confounding and population differences probably explain part of the gap between estimates.³⁰ Variant mismatch and immune history add a biological source of variation: BQ.1.1 and XBB.1 substantially escaped neutralization after a BA.5-bivalent dose, while XBB exposure often recalled B cells recognizing epitopes shared with the ancestral strain and recruited few truly XBB-specific clones in some people.^{31,32}

The outcome-specific declines and their supported, but not quantitatively partitioned, explanations are separated in Figure 2. The same systemic-versus-airway distinction seen in immune memory helps explain why protection from infection is less durable than protection from severe disease, but no single mechanism accounts for the full decline.

SUPPORTED · NOT QUANTITATIVELY PARTITIONED

-  **Circulating memory**
B and T cells can limit progression after infection.
-  **Airway gap**
Injected vaccine elicits weaker local airway immunity.
-  **Variant escape**
Changed antigen can evade previously matched antibodies.

Equal row size does not encode magnitude · CI = confidence interval · Evidence preceded widespread Omicron

Figure 2. Protection wanes faster against infection than severe disease. Average decline from month 1 to month 6 was 21.0 percentage points against infection and 10.0 points against severe disease in a pre-Omicron synthesis. Equal row sizes avoid a false magnitude encoding; airway immunity, immune memory and variant escape are supported explanations, but the literature does not assign a percentage to each.^{27,19,32,29}

07 Myocarditis risk is rare and concentrated

Large surveillance studies found that serious adverse outcomes were uncommon overall while identifying a specific myocarditis pattern. Active surveillance of **11.8 million doses** found no prespecified serious-outcome signal and anaphylaxis at about **5 per million doses**. A separate matched national study found that most examined events were not elevated but estimated 2.7 excess myocarditis cases per 100,000 vaccinated people.^{33,34} A 99-million-person study reproduced myocarditis and pericarditis signals, although comparisons depend on background rates. In adolescents, pooled myopericarditis was **43.5 per million doses** (95% CI 30.8–61.6): 66.0 in males versus 10.1 in females, and 60.4 after dose 2 versus 16.6 after dose 1.^{35,36}

Product and dose further concentrate risk. Nordic registers estimated **55.5 excess cases per million** after BNT162b2 dose 2 and **183.9 per million** after mRNA-1273 dose 2 in males aged 16–24. Booster risks were lower, at 8.6 and 19.5 per million young males respectively, but remained elevated.^{37,38} Infection is not a uniform comparator: across all ages it carried higher myocarditis risk, but among men younger than 40 the estimated excess after mRNA-1273 dose 2 was 97 per million, compared with 16 per million after a positive test.³⁹

The valid within-panel comparisons and incompatible cross-panel denominators are shown in Figure 3. The safety evidence repeats the review's wider pattern: a platform-wide average can obscure important formulation and population differences.

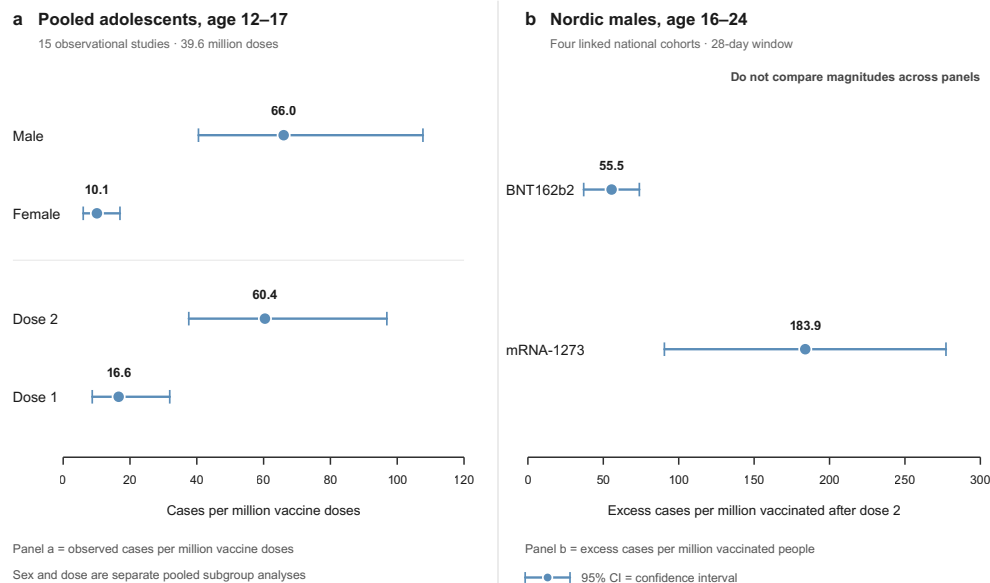


Figure 3. Myocarditis risk is concentrated rather than uniform. Panel a separates pooled adolescent rates by sex and dose; panel b separates Nordic excess events in males aged 16–24 by vaccine product. Populations, denominators and axes differ, so comparisons are valid within a panel but marker positions cannot be compared across panels.^{36,37,39,35}

08 Manufacturing is fast but formulation-specific

Large-scale production couples in-vitro transcription to DNA removal, purification, particle mixing, sterile filtration and product characterization. Double-stranded RNA control, mixing reproducibility and scale-up remain bottlenecks.^{40,3} Freeze-drying preserved one mRNA-LNP formulation for 12 weeks at room temperature and 24 weeks at 4 °C in preclinical tests. Continuous drying maintained transfection for at least 12 weeks at 4, 22 and 37 °C, but buffer, cryoprotectant and the ionizable-lipid:mRNA ratio determined whether a formulation survived drying.^{41,42,43}

Lipid chemistry also matters during liquid storage. Piperidine-based ionizable lipids reduced reactive aldehydes, RNA-lipid adducts and functional loss at refrigeration temperature. These studies identify possible routes to thermostability, not a universal room-temperature shelf life.⁴⁴ Rapid sequence exchange does not remove the need to validate the complete formulation and process.

09 New RNA formats differ in clinical maturity

Self-amplifying RNA carries replication machinery that copies the antigen template inside cells. A 5-µg ARCT-154 regimen showed 56.6% efficacy (95% CI 48.7–63.3) against any Delta-era COVID-19 and 95.3% (80.5–98.9) against severe disease. These results do not establish superiority across variants or over time.^{45,1} As a fourth-dose booster (n=828), ARCT-154 produced non-inferior ancestral-strain neutralization and higher BA.4/5 neutralization than BNT162b2 at day 28. The trial measured immune responses, not disease prevention or durability.⁴⁶

Circular RNA may extend expression without a replicase. An optimized Zika circular-RNA vaccine produced durable single-dose protection without obvious dengue enhancement in mice, but human safety and efficacy remain untested.⁴⁷ Self-amplification has therefore crossed into clinical efficacy testing, whereas circular RNA remains a preclinical prospect.

10 Conclusion

The cross-cutting pattern is that mRNA is programmable, but the biological product is the whole sequence-formulation-host system. Cytoplasmic delivery and chemical tuning make strong antigen expression possible; lymph nodes convert that brief signal into durable systemic memory; and clinical trials show that the platform can protect against more than one pathogen. The same evidence also defines its limits. Airway immunity is weaker than blood memory; protection from infection declines before protection from severe disease; antigenic drift can outpace the match; myocarditis risk is concentrated in particular groups and products; and stability gains do not automatically transfer between formulations. Self-amplifying RNA has clinical evidence, while circular RNA still requires human testing. Future progress therefore depends less on proving that mRNA can work than on optimizing each linked component and measuring performance in the population, pathogen and time horizon for which a vaccine is intended.

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